Multi-source Heterogeneous Data Fusion

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***Abstract*—As the exponential growth of data in internet era, there comes the big data era. Big data fusion creates huge values that makes it a research hotspot. However, in big data era, data shows characters of large volume, velocity, veracity and especially variety which is also called heterogeneity. Multiple different sources of data lead to data heterogeneity. Multi-source heterogeneous data brings opportunities and challenges to big data fusion. This paper introduces big data fusion and methods for heterogeneous data fusion, especially focus on the application of deep learning methods in multi-source heterogeneous data fusion. Challenges of dealing with multi-source heterogeneous data fusion is also discussed.**

Keywords-big data; heterogeneous data; data fusion; heterogeneous data fusion; deep learning

I. INTRODUCTION

With the rapid development of modern information technology and network technology, data sources are becoming more and more widespread and data types are getting richer and richer. The characteristics of big data, such as many data sources, diverse data structures, high data dimensions and fast growth, make the traditional data storage and data mining technologies face technological innovations [1-3]. Heterogeneity of data structure appears in structured data, semi-structured data and unstructured data, including relational data, spatial data, temporal data, text data, image data and audio and video data. The traditional multi-source data fusion mainly aims at the fusion of structured data such as sensor data. Different types of data are represented by different data feature. For example, text data is generally represented by discrete word vector features, whereas images are represented by pixel features of the image, including global and local features. The heterogeneity of data leads to the difference between the eigenvectors that represent the data, which becomes a gulf between multi-source heterogeneous data association, crossover and integration. Multi-source heterogeneous data fusion to meet the new features of the era of big data, data association analysis as a new challenge, has become a research hotspot [4-5]. Only by making full use of multi-source data and giving full play to the complementary nature of multi-source heterogeneous data can we conduct more thorough and thorough data analysis and obtain more valuable analysis results.

II. CHARACTERISTICS OF BIG DATA

A. 4V's Characters of Big Data

With the rapid popularization and development of the Internet, the data of the Internet has shown an exponential growth all over the world at an alarming rate, thereby leading the era of big data. Big data can be defined by its four characteristics, i.e., large volume, large velocity, large veracity and large variety, which is usually called 4V's model [6–8]. The most remarkable characteristic of big data is large-volume that implies an explosive in the data amount. For example, Flicker generates about 3.6 TB data and Google processes about 20,000 TB data every day. The National Security Agency reports that approximately 1.8 PB data is gathered on the Internet every day. Large velocity argues that big data is generating at an alarming rate and requires to be processed in real time. The real time analysis of big data is crucial for e-commerce to provide the online services. Another important characteristic of big data is large veracity that refers to the existence of a huge number of noisy objects, incomplete objects, inaccurate objects, imprecise objects and redundant objects [9]. The most distinctive characteristic of big data is large variety, which can also be understood as heterogeneity. It indicates the different types of data formats including text, images, videos, graphics, and so on. Most of the traditional data is in the structured format and it is easily stored in the two-dimensional tables. However, more than 75% of big data is unstructured. Typical unstructured data is multimedia data collected from the Internet and mobile devices [10].

Because of the variety of data acquisition devices, the acquired data are also in types with heterogeneity. Heterogeneity is one of major features of big data and heterogeneous data result in problems in data integration and Big Data analytics.

B. Multi-source Heterogeneous Data

Multi-source heterogeneous is similar to multi-modal, but contains more data types. In the field of information, modal can be understood as the existence of data formats, such as text, audio, image, video and other formats. Symbiotic or co-occurrence of a variety of single-modal information collectively called the so-called multi-modal. For example, video as a multimedia medium can be decomposed into multiple single-modal modes such as dynamic images, dynamic voices and dynamic texts. Images in the Internet are co-occurring with corresponding explanatory texts. As can be seen from the definition of multimodal, multimodal data is heterogeneous, but all belong to unstructured data. And multi-source heterogeneous data contains structured, semi-structured and unstructured data, covering the multi-modal data types. Therefore, the difficulty of multi-source heterogeneous data fusion can't be underestimated.

Different sources of data provide different aspects of the target object, so multi-source heterogeneous data can compensate for the shortcomings of incomplete data of a single data source, making the target information more adequate. By eliminating the gap between heterogeneous data and fusion of various data sources for correlation analysis, the data could emerge more valuable new information, to achieve the "1+1> 2" effect [11, 12]. Therefore, heterogeneous data poses opportunities and challenges on big data analysis.

III. BIG DATA FUSION

Data fusion is a prerequisite for quality assurance and analytic mining of integrated data. However, data fusion as a whole is a black box process for the user, which makes the data fusion process lacking in interpretability and debuggability [1].

Data fusion is generally divided into three levels: data layer fusion, feature layer fusion and decision layer fusion. Data layer fusion is the direct integration of data on the original data and analysis, is a low-level integration. The feature-level fusion is a fusion of the middle level. Firstly, the feature extraction of the original data is performed, and then the feature data is analyzed and processed synthetically. It achieves considerable information compression and facilitates real-time processing. At the decision-making level, the data are processed separately from different sources, including pretreatment, feature extraction, identification or discrimination, respectively, and the preliminary conclusions are obtained respectively. Then, the decision-making fusion decision is made through the correlation processing, and finally the joint inference result is obtained. The integration of data layers requires that the data be homogeneous and isomorphic while the structure of the multi-source heterogeneous data is heterogeneous, for example, an RGB matrix of 24 bits in the image, the audio is a string of 16-bit sound pressure values and the Chinese text is 16 or 24 Chinese character encoding. Heterogeneous data structure is also extracted heterogeneity. Therefore, how to make these heterogeneous data work together to accomplish an objective, such as identifying or retrieving tasks, through an effective fusion method is the primary problem to be solved by multi-source heterogeneous data fusion.

IV. MULTI-SOURCE HETEROGENEOUS DATA FUSION

Traditional multi-source information fusion is a method of information processing for sensors or multi-source systems. It processes the measurement information obtained from multiple sources and uses methods such as information association, information integration, and filtering to improve the estimation accuracy of the target state and other features, and ultimately to correct the situation, threats and their importance assessment [13, 14]. Multi-source information fusion originated in military applications in the latter part of World War II. It began in the 1970s and has been widely used in the fields of modern military affairs, livelihood, industry, transportation and finance, and has played a great role especially in the military field. Its processing object is obtained from various sources of measurement information, that is, the sensor data. It is simply structured scalar data which is also known as multi-sensor information fusion technology.

Heterogeneity is one of the essential features of big data. The data from different and various sources inherently possess many different types and representation forms, and it may be interconnected, interrelated, and represented inconsistently. Heterogeneity of big data also means that it is an obligation to acquire and deal with structured, semi-structured, and even entirely unstructured data simultaneously [15]. Big data relates large-volume and complex datasets with multiple independent sources. The analysis of big data can be troublesome because it often involves the collection and storage of mixed data based on different patterns or rules (heterogeneous mixture data) [16]. It is important to increase context awareness. For example, existing data in manufacturing has no relation to the context about users' history, schedule, habits, tasks, and location, etc. In the context of big data, contextualisation can be an attractive paradigm to combine heterogeneous data streams for improving the quality of a mining process or classifier. In addition, context awareness has demonstrated to be useful in reducing resource consumption by concentrating big data generation processes (e.g., the monitoring of real-world situations via cyber-physical systems) only on the sources that are expected to be the most promising ones depending on currently applicable (and often application-specific) context [17].

A. *Methods for Multi-Source Heterogeneous Data Fusion*

Zheng Yu of Microsoft Research Institute divided heterogeneous data fusion methods into three types [5]: 1) stage-based data fusion method, 2) feature-level data fusion method, and 3) semantic meaning-based data fusion method. Semantic-based fusion methods fall into four categories: 1) multi-view-based methods, 2) similarity-based methods, 3) probabilistic-dependency-based methods, 4) transfer learning-based methods.

Stage-based data fusion method refers to that in the process of staged data mining analysis, different stages of the use of different data to achieve the analysis purpose of this stage, so that multi-source heterogeneous data could fusion in the entire data mining analysis process. Such a fusion method makes the connection between different databases not close, nor does it have uniform requirements on the structure and modalities of the data used in each stage. [18] first detected traffic anomalies using GPS trajectory data and road network data, and identified social media information (such as Twitter) related to the location of traffic anomalies, and then analyzed the specific event content of traffic anomalies based on the retrieved relevant media information. In this process, heterogeneous data from different sources is used in both the detection of traffic anomalies and the analysis of the contents of specific traffic anomalies, without cross-over. Reference [19] proposed a multi-level multi-source heterogeneous data fusion method based on the problem of multi-source heterogeneous sensing information processing in the Internet of Things (IoT). Based on the application of target location tracking based on wireless signal, video and depth perception data, Multi-source heterogeneous data processing, feature representation and data fusion methods, by mining wireless signals, video and depth and other sources of heterogeneous data inherent relevance, to achieve the effective use of multi-source heterogeneous data valuable information.

From an overall perspective, the stage-based data fusion method has the original intention of making multi-source heterogeneous data to accomplish one goal together. However, at each stage, there is no interaction between heterogeneous data in the phase-based fusion method, losing the advantages of the complementary between heterogeneous data. Stage-based data fusion method does not across the semantic gap between heterogeneous data to achieve a truly intrinsic data fusion.

Feature-based fusion is another method of multi-source heterogeneous data fusion. In the process of feature layer fusion, data fusion is performed in the middle layer of data processing. Firstly, the features of each heterogeneous data are extracted and then the data is analyzed and processed to form a joint feature matrix or vector of multi-source heterogeneous data. The quality of the feature extracted from the feature-based fusion method and the method of fusion of the heterogeneous features all have a decisive influence on the effect of the method. The original feature-based fusion method is relatively rough, directly concatenate the features of multi-source heterogeneous information to form a new feature vector, and then used for clustering or classification analysis. Such feature fusion method ignores the redundancy among multi-source heterogeneous information features, and the correlation and the effect are not ideal [20]. The emergence of deep learning has improved the feature of multi-source heterogeneous data and made great progress in feature fusion effect, which makes the study of multi-source heterogeneous data fusion a big step forward.

B. *Deep Learning for Heterogeneous Data Fusion*

The early stage of deep learning is mainly used in the field of computer vision such as image classification and recognition. The mature deep learning model is more suitable for single data source learning [20], and the research on multi-source heterogeneous data fusion learning is not yet mature. Faced with many new data sources, many data structure types and high data dimension, data mining analysis also poses new challenges: How to utilize the association, cross-over and fusion of multi-source heterogeneous data to maximize the value of big data [1]. Fusion of multiple sources of heterogeneous data for a comprehensive analysis, the use of multi-source heterogeneous data contained in the complementary information, you can more fully and more fully dig out the target information. Combining deep learning and multi-source heterogeneous data fusion is a combination of cutting-edge technology and the characteristics of the times, with the latest technology to solve the latest challenges [4,21,22]. One of the most typical features of big data is the diversification of data formats. Data heterogeneity inevitably leads to differences in data eigenvectors. Eigenvector fusion is an important part of multi-source heterogeneous data fusion. Eliminating the differences between feature vectors and merging, it is one step closer to the realization of multi-source heterogeneous information fusion. Deep learning models learn the hierarchical features of data through unsupervised training and monitoring of test set parameters, and have achieved significant results in feature learning and extraction, as well as preliminary studies on the fusion of heterogeneous features [23-26].

From the feedforward neural network such as perceptron, multi-layer perceptron to other feedback neural networks such as RBM, DBM and CNN, depth learning has more and more advantages in feature learning performance. The research and application of multi-source heterogeneous data fusion through deep learning feature fusion are also more and more widely. Reference [27] integrated the deep learning eigenvectors of different dimensions into a unified and more descriptive eigenspace by integrating the face information of different views, and then fuses the features to realize the face recognition based on the deep heterogeneous features. Reference [28] extracted the eigenvectors of the audio data and video data of the target object through the deep learning model, and then combined the audio and video eigenvectors to form the integrated eigenvectors of the target object for the next step of identification and analysis. Through non-linear fusion of three sources of human heterogeneous information features through deep learning model, [29] more accurately estimated body posture. Reference [30] set up two Boltzmann machines to study the text and image features, respectively, and then the two features were combined into a new feature vector to be used as the input feature of the SVM classifier to complete the classification task. As a new hot spot in the field of deep learning, multi-source heterogeneous data fusion based on deep learning needs further research and exploration to make more discoveries and progress.

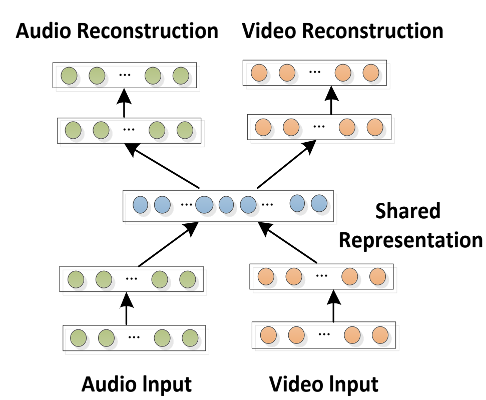

Figure 1. Bimodal Deep Autoencoder in [28].

TABLE I. MULTI-MODAL LEARNING SETTINGS IN [28]

	Feature Learning	Supervised Training	Testing
Classic Deep Learning	Audio	Audio	Audio
	Video	Video	Video
Multimodal Fusion	Audio +Video	Audio +Video	Audio +Video
Cross Modality Learning	Audio +Video	Audio	Audio
	Audio +Video	Video	Video
Shared Representation Learning	Audio +Video	Audio	Video
	Audio +Video	Video	Audio

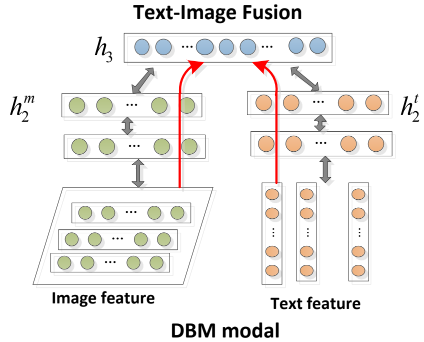

Figure 2. DBM modal for text-image fusion in [30].

V. CONCLUSION

Big data has become a distinctive feature of today's era, the analysis of big data has become a research hotspot. As one of the characteristics of big data, multi-source heterogeneity has brought opportunities and challenges for big data analysis and has great research value. Multi-source heterogeneity of big data means dealing with structured, semi-structured, and unstructured data simultaneously. Traditional data fusion methods have limitations in multi-source heterogeneous data fusion. Much research and many new fusion methods have appeared to solve this problem. Deep learning is capable of the analysis and learning of massive amounts of unsupervised data. What's more, deep learning could fusion the fusion the feature of multi-source heterogeneous data which is of considerable effect. However, there are still many problems to be solved in multi-source heterogeneous data fusion, such as how to solve heterogeneity problems among data universally, how to achieve better integration effect, how to stand at the height of artificial intelligence to realize multi-source heterogeneous data fusion at semantics level, and so on. We still have a lot to do.

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